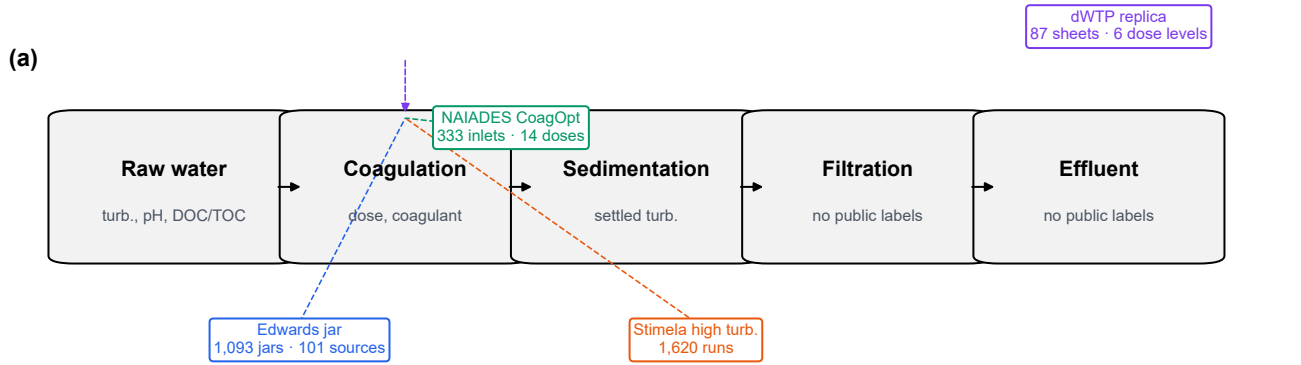


Fig. 1. Scope of the case study. (a) Coagulation–sedimentation chain with four datasets mapped to observable stages (dashed); filtration and effluent stages lack labels in the public benchmarks used here. (b) Three lenses: grouped CV, label identifiability (Stimela: flocculated turbidity = inlet turbidity × k(dose)), and supported vs unsupported ML claims.